- **Machine Learning**: we have briefly introduced the field and seen that it includes three main branches: *supervised learning, unsupervised learning, reinforcement learning*.
- **Clustering:** is the main application of unsupervised learning, used to make sense of unlabeled data, discovering patterns in raw data.
- **Applications:** some applications of clustering are marketing segmentation, noise detection (f.i. in signals) and data preprocessing.
- **K-means**: the oldest clustering algorithm, it's the easiest to implement, but has also limitations in the shape of the clusters: it can only spot convex clusters, and can't handle non-linearly-separable data.
- **DBSCAN**: a different approach to clustering, identifying groups based on the density of the points; it can handle any shape and non-linearly separable data, but it doesn't work well with dataset with areas of heterogenous density; moreover, it's hard and not intuitive to choose the best values for the algorithm hyper-parameters.
- **OPTICS**: a newer method, based on DBSCAN, that builds a hierarchical clustering and allows to handle datasets with varying density; it also makes it easier to choose the values of the parameters.
- **Metrics**: to assess the quality of a clustering for a dataset, we can use evaluation metrics like intra-cluster distance, inter-cluster distance or total cohesion.
- **Elbow method**: a tool that provides graphical feedback to decide the best values for the algorithms' hyper-parameters; we have seen, in particular, how to use it